SPRiF: Spectral Memory and Scalar Reset for Spiking Neurons

Author information to be completed

***Abstract*—A network may need an early cue long after the neurons that received it have fired. In a leaky integrate-and-fire neuron, the cue and each subsequent reset contribute to the same voltage trace. We introduce SPRiF, a five-state neuron that stores input current in a spectral memory and resets the membrane that reads it. One real decay and a damped rotation provide the temporal basis. Two learned projections drive the membrane, one immediately and one through an auxiliary filter delayed by one step. We derive the membrane drive in terms of the input history and identify the additional spike-history term produced by a reset along the memory input direction. We evaluate SPRiF on five temporal benchmarks and test reset placement and mode organization with matched controls on two of them. Preserving memory at reset gave mean accuracy gains of 3.37 percentage points on permuted sequential MNIST and 4.56 on Spiking Heidelberg Digits over the memory-reset control. On delayed match-to-sample, SPRiF retained 99.42% mean accuracy under 40 forced-spike timesteps after clean training. Training with up to eight intervention timesteps yielded 99.33% under the same stress, averaged over five delays including an unseen 2500-ms delay.**

***Index Terms*—Spiking neural networks, temporal processing, spectral memory, neuron dynamics, scalar reset.**

I. Introduction

In a delayed decision, the first cue may be gone by the time the second arrives. Spikes emitted between the cues serve two roles in a recurrent spiking neural network (SNN). They reset the membranes that fired and enter the recurrent current on the next step. In the leaky integrate-and-fire (LIF) model, an input pulse and a subsequent reset leave traces in the same membrane voltage, both governed by the same decay. The signal available at the second cue therefore depends on the intervening spikes as well as the first cue.

A neuron with several state variables can assign these roles differently. Adaptive neurons retain a history of activity [1], [2], heterogeneous neurons and synapses distribute temporal integration across time scales [3], [4], resonant dynamics produce lag-dependent responses [5]–[7], and coupled compartments distribute integration across states [8], [9]. PMSN separates linear hidden compartments from a firing compartment [10], while DMP-SNN maintains a memory shared across a layer [11]. Reset itself can also be softened, adapted or removed [12]–[14]. These developments motivate a joint question about the organization of a fixed number of states within one neuron: which temporal responses can the membrane read, and how does a spike change those responses?

SPRiF assigns three coordinates to a spectral memory, one to a filter and one to the membrane. A pulse excites a decaying trace and a damped rotation. The membrane reads a projection of the current spectral state and a filtered projection from the preceding step. These two paths allow the neuron to change the weight of older inputs while holding the instantaneous response fixed. The local reset acts on the membrane alone. For a specified current sequence, we derive the kernel that maps input to membrane drive and identify the extra spike-history term introduced when reset also acts on the spectral state.

We evaluate SPRiF on five temporal benchmarks and test the two design choices on permuted sequential MNIST (PS-MNIST) and Spiking Heidelberg Digits (SHD). Two controls retain the same five states, eleven learned local coefficients and layer widths. One adds a reset to spectral memory. The other replaces the spectral block with three driven real decays. A second experiment, Spike-Intervention Delayed Match-to-Sample (SI-DMS), forces spikes between two cues. It tests whether the full recurrent network can still compare the cues when its spike history changes.

II. Related Work

A. Temporal Dynamics in Spiking Neurons

Adaptive neurons retain activity in a variable that influences later firing. ASRNN combines adaptive spiking dynamics with recurrent processing for time-domain classification [1], while the adLIF family, including the SE-adLIF comparator in Table I, exploits adaptation for spatiotemporal processing [2]. SNN-SFA uses spike-frequency adaptation to retain information through a slowly varying firing threshold [15]. Earlier spikes leave a trace in neuronal excitability that can support the subsequent recall of temporally dispersed inputs. RadLIF instead combines recurrent connections with an adaptive current driven by subthreshold voltage and spike-triggered increments [16]. These models illustrate two ways to carry activity beyond the membrane time constant: changing the threshold for future firing or changing the current that opposes membrane integration.

DA-LIF introduces independently learnable decays for spatial and temporal tuning [17]. Temporal diversity can also be distributed across neurons: Heterogeneous SNN distributes neuronal parameters to support different integration times [3], and bimodal time-scale networks explicitly combine populations with short and long time constants [18]. HetSyn instead assigns time constants to individual synapses, placing temporal integration in synaptic currents [4]. Conductance-based models obtain multiple time scales through fast, slow and ultra-slow conductances [19]. These approaches broaden the temporal dynamics available to a network by changing the variables that retain activity and the rates at which they evolve. For SPRiF, the design question also concerns the shape of the retained response. A real decay assigns a fixed-sign trace to an isolated pulse, whereas a rotational pair gives the readout access to its changing phase. Combining these components provides both persistence and phase-dependent weighting within one neuron.

Resonate-and-fire neurons use coupled states whose phase affects the response to a past input [5]. BRF develops balanced resonate-and-fire dynamics for trainable spiking networks [6]. PRF combines resonant membrane dynamics with a decoupled reset construction for parallel sequence processing [20].

Compartmental models offer another organization: Dendrify provides reduced models with dendritic integration and local spikes [21], TC-LIF couples dendritic and somatic states [8], and DH-SNN assigns heterogeneous time constants to dendritic branches [9]. A close relative of SPRiF is D-RF, whose resonant dendrites feed a somatic readout and whose threshold depends on spike history [7]. Its dendrites also evolve without subtractive output reset. SPRiF gives the membrane two learned projections of its spectral states, with the second acting through an auxiliary filter one step later. This readout structure separates the immediate response from part of the delayed response: changing the filtered projection alters later membrane drive without changing its value at the arrival of an input pulse. The temporal kernel makes this distinction explicit.

State-space models motivate both the parameterization of spiking dynamics and their organization within a sequence model. SiLIF transfers learnable discretization and logarithmic parameterization to spiking neurons, with a complex-state variant that supports oscillatory regimes [22]. SpikingSSMs combine state-space sequence processing with sparse spiking outputs [23]. SPRiF specifies a real pole and a complex-conjugate pair directly, then learns how their states reach the membrane through immediate and delayed projections.

B. Reset Mechanisms and Spike History

PMSN separates the dynamics of its hidden compartments from discharge at its output [10]. Unrolling its diagonalized hidden transition gives a convolution from injected current to the drive at the output compartment. Its parallel output update requires a separate construction, temporal-decoupled reset, which operates on rectified current and discharges integer multiples of the threshold. SPRiF uses a leaky membrane with signed drive and one threshold subtraction per binary spike. This choice gives a closed form for the drive to the membrane while retaining a sequential firing decision.

Reset design determines how firing changes the state available at later timesteps. Guo *et al.* use a soft reset and membrane-potential rectification to reduce information loss [12]; AR-LIF adapts reset through the relation between input and output [13]. RPLIF models refractory effects through spike-triggered threshold dynamics [24], while Zhang analyzes reset and refractory periods from the perspective of binary-activated recurrent networks [25]. PSN removes reset to obtain parallelizable temporal computations [14]. SPRiF retains a subtractive membrane reset and analyzes which additional history term appears when that reset is extended to spectral memory.

NiSNN-A writes LIF voltage as the difference between accumulated input and accumulated reset [26]. Its NiLIF update estimates the unknown spike train from the input-only accumulation to obtain a noniterative computation. In SPRiF, spikes are generated by thresholding the actual membrane candidate at each step. The local response to a realized spike can then be derived from the state equations.

C. Gating and Memory Organization

Gating provides another means of controlling how information enters and persists in a spiking neuron. GLIF learns gating factors that combine alternative biological features of membrane leakage, input integration and spike initiation [27]. Its learned gates and membrane parameters enlarge the family of neuronal dynamics that a network can express. DGN introduces activity-dependent membrane conductance as a dynamic gate [28]. This mechanism selectively filters incoming signals and suppresses disturbances, linking neuronal activity to the strength of subsequent input transmission. Both approaches make the regulation of information flow an explicit part of neuron design. SPRiF instead specifies a linear spectral memory and learns its projections into the membrane, with the threshold nonlinearity applied at the output.

Temporal coordination can also be studied at the network level. MPS-SNN views successive timesteps as temporal subnetworks with shared architectures and weights [29]. It uses membrane-potential smoothing and guidance between adjacent temporal subnetworks to reduce inconsistencies in their states and outputs. The smoothing mechanism also supports the propagation of information and gradients through time. This perspective emphasizes consistency across successive network evaluations. The SPRiF analysis focuses on the response retained inside each neuron: how an input pulse is transformed over time and how an intervening spike changes the drive available to later decisions.

DMP-SNN makes a different architectural choice by sharing memory within a layer [11]. Layer-level sharing and neuron-local state organization provide different ways to allocate a network's memory resources. SPRiF addresses the latter by specifying how five local states receive input, communicate with the membrane and respond to firing. Its matched controls keep the number of states and learned dynamics coefficients fixed, so the comparison focuses on their temporal modes and reset placement.

Together, these studies motivate separating the temporal response carried by a state from the changes imposed on it by firing. SPRiF examines both within one neuron: its spectral block determines the basis available to the membrane, and its reset rule determines whether local spikes directly modify that basis state. This distinction also separates local memory reset from recurrent feedback. A spike can leave the memory coordinates unchanged at discharge and still alter their future trajectory through the recurrent input. The theoretical analysis distinguishes these two paths, while SI-DMS evaluates their combined effect on delayed decisions.

III. SPRiF Neuron Model and Analysis

A. Neuron Formulation

To fix the timing convention, let v_t be a LIF membrane immediately after the current reset. Its candidate, spike, and

stored voltage are

$$\widetilde{v}_t = \beta v_{t-1} + I_t, \qquad z_t = H(\widetilde{v}_t - \theta), \qquad v_t = \widetilde{v}_t - \theta z_t, \quad (1)$$

where $0 < \beta < 1$, $\theta > 0$, and $H(a) = 1$ for $a \geq 0$ and zero otherwise. Unrolling gives

$$v_t = \beta^{t+1} v_{-1} + \sum_{j=0}^{t} \beta^{t-j} I_j - \theta \sum_{j=0}^{t} \beta^{t-j} z_j. \quad (2)$$

A unit of current at step j contributes β^{t-j} at step t. A spike at the same step contributes $-\theta\beta^{t-j}$. The two contributions differ in sign and scale but have the same time course. SPRiF retains membrane reset and gives the membrane access to a second history, a filtered record of the input current whose coordinates are preserved at discharge.

For an input vector $\mathbf{s}_t$ and a recurrent spiking layer, the current vector is

$$\mathbf{I}_t = W_{\text{in}} \mathbf{s}_t + W_{\text{rec}} \mathbf{z}_{t-1} + \mathbf{b}. \quad (3)$$

The recurrent term is omitted for a feedforward layer. The equations below concern one neuron and suppress its index. Its current I_t is the corresponding component of $\mathbf{I}_t$, and it stores $\mathbf{x}_t \in \mathbb{R}^3$, $h_t \in \mathbb{R}$, and $u_t^0 \in \mathbb{R}$:

$$\begin{cases} \mathbf{x}_t = A\mathbf{x}_{t-1} + BI_t, \\ h_t = \eta_1 h_{t-1} + \bar{\mathbf{g}}_1^\top \mathbf{x}_t, \\ \widetilde{u}_t^0 = \eta_0 u_{t-1}^0 + h_{t-1} + \mathbf{g}_0^\top \mathbf{x}_t, \\ z_t = H(\widetilde{u}_t^0 - \theta), \\ u_t^0 = \widetilde{u}_t^0 - \theta z_t. \end{cases} \quad (4)$$

The transition and input vector are

$$A = \begin{bmatrix} \alpha & 0 & 0 \\ 0 & \rho\cos\omega & -\rho\sin\omega \\ 0 & \rho\sin\omega & \rho\cos\omega \end{bmatrix}, \qquad B = \begin{bmatrix} 1-\alpha \\ 1-\rho \\ 0 \end{bmatrix}. \quad (5)$$

The two directly learned readouts form

$$\bar{G} = \begin{bmatrix} \mathbf{g}_0^\top \\ \bar{\mathbf{g}}_1^\top \end{bmatrix} \in \mathbb{R}^{2\times 3}. \quad (6)$$

For finite raw parameters, the constraints are enforced by

$$\begin{aligned} &\alpha = \sigma(\alpha_{\text{raw}}), \qquad \rho = \sigma(\rho_{\text{raw}}), \\ &\omega = \pi\sigma(\omega_{\text{raw}}), \quad \eta_i = \sigma(\eta_{i,\text{raw}}), \quad i \in \{0,1\}. \end{aligned} \quad (7)$$

Here σ is the logistic sigmoid. Thus $\alpha, \rho, \eta_i \in (0,1)$ and $\omega \in (0,\pi)$. The six entries of $\bar{G}$ are unconstrained real coefficients. The threshold is a fixed task setting.

Figure 1 contrasts the state organization of LIF and SPRiF. In SPRiF, two routes connect $\mathbf{x}_t$ to firing. The direct projection enters $\widetilde{u}_t^0$ immediately. The second projection updates h_t, which first enters the membrane at $t+1$. At the current step the membrane reads h_{t-1}. All intrinsic states start at zero for an independent sequence.

The filter-to-membrane coupling in Eq. (4) has unit gain. A separate constant gain κ would be redundant with the unconstrained input projection of the filter. For a filter $u_t^1 = \eta_1 u_{t-1}^1 + \mathbf{g}_1^\top \mathbf{x}_t$ read out as κu_{t-1}^1, set

$$h_t = \kappa u_t^1, \qquad \bar{\mathbf{g}}_1 = \kappa \mathbf{g}_1, \qquad h_{-1} = \kappa u_{-1}^1. \quad (8)$$

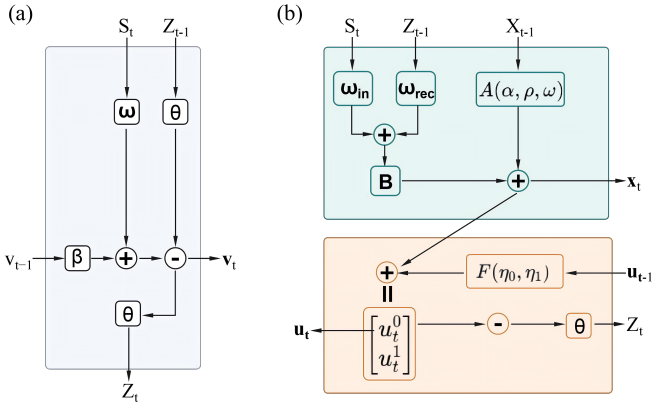


Fig. 1. State organization of LIF (left) and SPRiF (right). LIF integrates input and reset in one membrane. SPRiF separates spectral memory from the auxiliary filter and membrane, with spike-triggered reset confined to the membrane.

Substitution gives the same forward membrane voltage and spikes for fixed coefficients, including at $\kappa = 0$. SPRiF learns $\bar{\mathbf{g}}_1$ directly. This leaves five decay or frequency parameters and six projection coefficients per neuron.

Each neuron stores (x^0, x^1, x^2, h, u^0), for five state values. It learns the five raw variables associated with $\alpha, \rho, \omega, \eta_0, \eta_1$ and six entries of $\bar{G}$, for eleven dynamics coefficients. Thresholds are fixed, and synaptic weights, biases, and output parameters are separate.

For a recurrent layer with input width m and hidden width n, a dense input matrix, dense recurrent matrix, and one current bias per neuron contribute

$$N_{\text{layer}} = mn + n^2 + n + 11n. \quad (9)$$

Complete network counts include the task readout and use the bias settings of the corresponding layer. A width-n layer stores $5n$ intrinsic state values in addition to its spike cache. Its local update takes $O(n)$ operations per step, while the dense synaptic products take $O(mn + n^2)$.

B. Temporal Kernel Analysis

The poles of the spectral transition are

$$\text{spec}(A) = \{\alpha, \rho e^{\text{i}\omega}, \rho e^{-\text{i}\omega}\}. \quad (10)$$

For a unit impulse $I_0 = 1$, zero state at $t = -1$, and zero later current,

$$A^k B = \begin{bmatrix} (1-\alpha)\alpha^k \\ (1-\rho)\rho^k \cos(k\omega) \\ (1-\rho)\rho^k \sin(k\omega) \end{bmatrix}, \qquad k \geq 0. \quad (11)$$

The first coordinate retains one sign. In the other two, ρ sets the decay per step and ω sets the phase advance. For example, at $\omega = \pi/2$ the cosine coordinate equals $1-\rho$ at lag zero and $-(1-\rho)\rho^2$ two steps later. A projection onto that coordinate can therefore give the same pulse opposite weights according to its age.

The auxiliary filter accumulates a second projection of this trajectory. For the same pulse its response is

$$h_k = \sum_{\ell=0}^{k} \eta_1^{k-\ell} \bar{\mathbf{g}}_1^\top A^\ell B. \quad (12)$$

Write the membrane drive as

$$d_t = \mathbf{g}_0^\top \mathbf{x}_t + h_{t-1}. \tag{13}$$

Substituting the spectral and filter responses gives the kernel

$$\begin{aligned} k_d[k] &= \mathbf{g}_0^\top A^k B \\ &\quad + \sum_{\ell=0}^{k-1} \eta_1^{k-1-\ell} \bar{\mathbf{g}}_1^\top A^\ell B, \end{aligned} \tag{14}$$

The sum is empty at $k = 0$. In particular, $k_d[0] = \mathbf{g}_0^\top B$ and $k_d[1] = \mathbf{g}_0^\top AB + \bar{\mathbf{g}}_1^\top B$. Adjusting the second projection can change the response one step after a pulse while leaving its onset unchanged. Writing ξ for a unit delay gives the transfer function

$$\mathcal{K}_d(\xi) = \left(\mathbf{g}_0^\top + \frac{\xi}{1 - \eta_1 \xi} \bar{\mathbf{g}}_1^\top \right) (I_3 - \xi A)^{-1} B. \tag{15}$$

For fixed coefficients and zero initial state, convolving k_d with the realized current sequence gives the exact drive. The membrane then integrates this drive and its own reset history. For any initial membrane voltage,

$$\begin{aligned} u_t^0 &= \eta_0^{t+1} u_{-1}^0 + \sum_{j=0}^{t} \eta_0^{t-j} d_j \\ &\quad - \theta \sum_{j=0}^{t} \eta_0^{t-j} z_j. \end{aligned} \tag{16}$$

Equation (16) assigns the membrane its own leak η_0. The drive that precedes this integration has the poles of the spectral block and the auxiliary filter, subject to cancellation by the learned projections.

The complex pair contributes the quadratic factor $1 - 2\rho\cos(\omega)\xi + \rho^2\xi^2$ to the denominator of Eq. (15). If this factor survives in the readout, a block containing only positive real decays and the same auxiliary filter cannot reproduce the kernel at all lags. Its denominator has only real linear factors. The distinction is an intrinsic oscillatory mode, since signed sums of real decays can themselves change sign. Section IV-C defines a control with three driven real modes.

C. Controllability and State Boundedness

Current enters two spectral coordinates directly (Eq. (5)). Rotation couples one of them to the third. The resulting three-dimensional state can be reached with a scalar current.

Proposition 1 (Driven spectral coordinates): For $\alpha, \rho \in (0, 1)$ and $\omega \in (0, \pi)$, the pair (A, B) in Eq. (5) is controllable.

Proof: Let $\mathcal{C} = [B, AB, A^2B]$. Expanding its determinant gives

$$\begin{aligned} \det \mathcal{C} &= (1 - \alpha)(1 - \rho)^2 \rho \sin\omega \\ &\quad \times (\alpha^2 - 2\alpha\rho\cos\omega + \rho^2). \end{aligned} \tag{17}$$

Every factor is positive, since the last factor equals $|\alpha - \rho e^{\mathrm{i}\omega}|^2$. Thus $\mathcal{C}$ has rank three. ∎

From zero state, three appropriately chosen scalar inputs can therefore reach any point in the spectral state space. Reachability concerns the state before projection. A learned readout can suppress a reachable mode, and a nearly singular controllability matrix can make some directions difficult to excite.

For fixed parameters and a specified current sequence, define $\mathbf{m}_t = [\mathbf{x}_t^\top, h_t]^\top$. The memory pathway is

$$\mathbf{m}_t = F\mathbf{m}_{t-1} + LI_t, \tag{18}$$

with

$$F = \begin{bmatrix} A & \mathbf{0} \\ \bar{\mathbf{g}}_1^\top A & \eta_1 \end{bmatrix}, \qquad L = \begin{bmatrix} B \\ \bar{\mathbf{g}}_1^\top B \end{bmatrix}. \tag{19}$$

The block-triangular structure gives

$$\mathrm{spec}(F) = \{\alpha, \rho e^{\mathrm{i}\omega}, \rho e^{-\mathrm{i}\omega}, \eta_1\}. \tag{20}$$

Each pole is strictly inside the unit disk. The learned projection into h_t can amplify a transient even though every mode decays.

Proposition 2 (Bounded local states): Assume fixed finite readouts, finite initial states, $\theta > 0$, the ranges in Eq. (7), and $|I_t| \le M$ for all $t \ge 0$. Then $\mathbf{x}_t$, h_t, and u_t^0 in Eq. (4) remain bounded.

Proof: Let $q = \max(\alpha, \rho) < 1$. The rotational part of A^k is a rotation scaled by ρ^k, so $\|A^k\|_2 = q^k$. Summing the geometric series for the forced spectral recurrence gives a uniform bound X on $\|\mathbf{x}_t\|_2$. Applying the same argument to the filter gives a bound H_* on $|h_t|$, where

$$\begin{aligned} X &= \|\mathbf{x}_{-1}\|_2 + \frac{\|B\|_2 M}{1 - q}, \\ H_* &= |h_{-1}| + \frac{\|\bar{\mathbf{g}}_1\|_2 X}{1 - \eta_1}. \end{aligned} \tag{21}$$

It follows that $|d_t| \le \|\mathbf{g}_0\|_2 X + H_*$. Since each spike subtracts at most θ, the expansion in Eq. (16) yields

$$|u_t^0| \le |u_{-1}^0| + \frac{\|\mathbf{g}_0\|_2 X + H_* + \theta}{1 - \eta_0}. \tag{22}$$

All three bounds are finite. ∎

For a prescribed current sequence, a perturbation within the memory and filter coordinates propagates as $\delta\mathbf{m}_{t+k} = F^k \delta\mathbf{m}_t$. Poles near the unit circle permit slow decay along this path. If a spike changes a subsequent current, an additional input-driven response must be included.

D. Reset Effects on Memory Dynamics

Use superscripts $-$ and $+$ for values immediately before and after the current local reset. Equation (4) gives

$$\mathbf{x}_t^+ = \mathbf{x}_t^-, \qquad h_t^+ = h_t^-, \qquad u_t^{0,+} = \widetilde{u}_t^0 - \theta z_t. \tag{23}$$

A spike changes stored voltage immediately. All four memory and filter coordinates have zero reset increment, so for a given current sequence the drive d_t contains no direct contribution from past local resets.

To see what changes when a spike also acts on memory, consider the control in Eq. (29). After both projections have read $\mathbf{x}_j^-$, a spike at j adds $-\theta\widehat{B}$ to the stored spectral state, where $\widehat{B} = B/\|B\|_2$. An input pulse of size $-\theta/\|B\|_2$ would produce the same spectral displacement. Its timing within the step would differ, however, because an input pulse enters h_j before the reset occurs. Under a unit pulse, that filter update

contributes $\eta_1^{k-1}\bar{\mathbf{g}}_1^\top B$ to the drive at lag $k \geq 1$. Removing this contribution from Eq. (14) and scaling by the signed pulse amplitude gives the kernel r from a spike to the membrane drive. The current drive has already been computed, so $r[0] = 0$. At later lags,

$$r[k] = -\frac{\theta}{\|B\|_2}\left(k_d[k] - \eta_1^{k-1}\bar{\mathbf{g}}_1^\top B\right), \quad k \geq 1. \tag{24}$$

For zero initial states, superposition in the memory and filter recurrences now gives

$$d_t = \sum_{k=0}^{t} k_d[k]\, I_{t-k} + \sum_{k=1}^{t} r[k]\, z_{t-k}. \tag{25}$$

The second sum is the direct contribution of memory reset in MEMORY_RESET5. In SPRiF its kernel is identically zero by Eq. (23). Each identity uses the model's own fixed coefficients and realized currents and spikes. The membrane in both models still receives its scalar reset.

A spike can also return through the network. Holding the external layer input fixed, a change in the spike vector at t changes the next recurrent current by

$$\delta\mathbf{I}_{t+1} = W_{\text{rec}}\, \delta\mathbf{z}_t. \tag{26}$$

This change enters the first sum of Eq. (25) through the same kernel as any other current. It is distinct from the direct memory reset in the second sum. SI-DMS tests the joint effect of overwriting membrane candidates, firing, reset and recurrent propagation on a delayed decision.

IV. Experimental Setup

A. Datasets and Network Architectures

S-MNIST presents each MNIST image [30] as 784 scalar inputs, normalized with mean 0.1307 and standard deviation 0.3081. PS-MNIST uses a pixel permutation generated from the run seed and shared between training and test. Variability across seeds on PS-MNIST therefore includes changes in pixel order. Both tasks use two recurrent layers of widths 64 and 210. At evaluation, the linear readout averages logits over the full sequence.

QTDB uses four event streams derived from two ECG channels of the QT Database [31]. The preprocessing code smooths each channel and thresholds its first differences at ± 0.03. The readout predicts six annotation groups at every timestep. Accuracy counts all positions in overlapping windows of length 1301 and stride 650. The generator lists record *sel16483* in both partitions. The record membership of the preprocessed files is unverified, which limits interpretation of this score as generalization to unseen records.

GSC uses 12 classes from Speech Commands [32]. Forty mel coefficients and their first and second differences give 120 features per frame. A 300-unit recurrent layer pools spikes over 101 frames for classification. SHD uses 700 cochlear input channels and 20 digit classes [33]. A 64-unit recurrent layer drives 20 nonrecurrent SPRiF output units. The readout sums the softmax of their pre-reset membrane values over indices $t > 10$.

B. Training and Evaluation Protocols

The forward pass uses a hard threshold. During backpropagation through time (BPTT), its derivative is replaced by the multi-Gaussian surrogate [1]

$$\begin{aligned}\psi(v) = \gamma\big[&a_0\mathcal{N}(v-\theta;0,\sigma_0^2)\\ &- a_1\mathcal{N}(v-\theta;\mu_1,\sigma_1^2)\\ &- a_1\mathcal{N}(v-\theta;-\mu_1,\sigma_1^2)\big],\end{aligned} \tag{27}$$

where $\mathcal{N}(\cdot;\mu,\sigma^2)$ is the normalized Gaussian density, $\sigma_0 = 0.5$, $\mu_1 = 0.5$, $\sigma_1 = 3.0$, $a_0 = 1.15$, $a_1 = 0.15$, and $\gamma = 0.5$. Gradients pass through the recurrent spike and the membrane reset within each BPTT segment.

The benchmark runs use Adam for 150 epochs on S-MNIST, PS-MNIST and GSC, 250 on QTDB and 100 on SHD. The initial learning rate is 0.01 except on GSC, where it is 0.005. Batch sizes are 512 for the MNIST tasks, 128 for QTDB and GSC, and 100 for SHD. On both sequential MNIST tasks, each sequence is split into segments of 262, 262 and 260 steps. Each segment receives an optimizer update. State values carry over, and gradients stop at the boundaries. Within a segment, gradients pass through the spike and reset using Eq. (27). QTDB scores a 31-unit recurrent layer at each timestep. GSC uses negative log likelihood on the pooled readout with class-weighted sampling.

For S-MNIST, PS-MNIST, QTDB and SHD, each run contributes its highest test accuracy across epochs. The reported test split therefore also serves for checkpoint selection. GSC contributes its highest validation accuracy. SI-DMS uses the final checkpoint after 2000 updates.

Each benchmark decay is initialized from a positive time constant in discrete steps:

$$q = \exp(-1/\tau_q), \qquad q_{\text{raw}} = \log\frac{q}{1-q}. \tag{28}$$

Frequency is parameterized analogously through ω/π. All intrinsic states start at zero. The two rows of $\bar{G}$ are optimized directly. The second row uses the product initialization in Eq. (8) on the benchmarks and starts at zero on SI-DMS. On the MNIST tasks the real-mode and filter time constants are sampled uniformly in steps, with the upper bound of the oscillatory time constant decreasing with frequency. QTDB, GSC and SHD use log-uniform time constants. The nominal $(\tau_\alpha, \tau_\rho, \tau_\eta)$ intervals are $([2,60],[2,40],[0.8,10])$ for both MNIST tasks, $([20,120],[4,30],[0.8,8])$ for QTDB, and $([10,80],[4,30],[0.8,8])$ for GSC and SHD. Here τ_η denotes the range for each filter or membrane decay. The respective frequency ranges for ω/π are $[0.04, 0.50]$, $[0.02, 0.20]$ and $[0.04, 0.40]$.

One training run is one replicate. We report mean accuracy and sample standard deviation across seeds $\{0, 1, 2, 42, 1111\}$. Published entries retain their original reporting conventions. In SI-DMS, each delay has equal weight within a seed. We report the mean Δ_{40}, equal to mean clean accuracy minus mean stressed accuracy.

C. Baselines and Ablation Models

TC-LIF [8] and BRF [6] were rerun on PS-MNIST and SHD. DMP-SNN [11] was rerun on PS-MNIST using its

TABLE I
PERFORMANCE ON S-MNIST, PS-MNIST, QTDB, GSC, AND SHD. REC: RECURRENT; *: OUR BASELINE RERUNS.

Dataset	Method	Net	Parameters (K)	Accuracy (%)
S-MNIST	GLIF$_{\text{NeurIPS, 2022}}$ (Yao et al. 2022) [27]	Rec	150	96.64
	ASRNN$_{\text{Nat. Mach. Intell., 2021}}$ (Yin, Corradi, and Bohté 2021) [1]	Rec	150	98.70
	DH-SNN$_{\text{Nat. Commun., 2024}}$ (Zheng et al. 2024) [9]	Rec	80	98.90
	BRF$_{\text{ICML, 2024}}$ (Higuchi et al. 2024) [6]	Rec	69	99.10
	TC-LIF$_{\text{AAAI, 2024}}$ (Zhang et al. 2024) [8]	Rec	150	99.20
	DMP-SNN$_{\text{Nat. Mach. Intell., 2026}}$ (Sun et al. 2026) [11]	DMP	73	99.20
	SPRiF (ours)	**Rec**	**67**	**99.28 ± 0.06**
PS-MNIST	GLIF$_{\text{NeurIPS, 2022}}$ (Yao et al. 2022) [27]	Rec	150	90.47
	ASRNN$_{\text{Nat. Mach. Intell., 2021}}$ (Yin, Corradi, and Bohté 2021) [1]	Rec	150	94.30
	DH-SNN$_{\text{Nat. Commun., 2024}}$ (Zheng et al. 2024) [9]	Rec	80	94.52
	BRF$_{\text{ICML, 2024}}$ (Higuchi et al. 2024) [6]	Rec	69	95.20 / 95.02 ±0.41*
	TC-LIF$_{\text{AAAI, 2024}}$ (Zhang et al. 2024) [8]	Rec	150	95.36 / 95.10 ±0.33*
	DMP-SNN$_{\text{Nat. Mach. Intell., 2026}}$ (Sun et al. 2026) [11]	DMP	61 / 67	95.50 / 95.41 ±0.52*
	SPRiF (ours)	**Rec**	**67**	**95.86 ± 0.14**
QTDB	ASRNN$_{\text{Nat. Mach. Intell., 2021}}$ (Yin, Corradi, and Bohté 2021) [1]	Rec	1.8	85.90
	DH-SNN$_{\text{Nat. Commun., 2024}}$ (Zheng et al. 2024) [9]	Rec	1.78	86.35
	BRF$_{\text{ICML, 2024}}$ (Higuchi et al. 2024) [6]	Rec	1.73	87.00
	SE-adLIF$_{\text{Nat. Commun., 2025}}$ (Baronig et al. 2025) [2]	Rec	1.8	86.88
	SPRiF (ours)	**Rec**	**1.7**	**88.43 ± 0.36**
GSC	SNN-SFA$_{\text{eLife, 2021}}$ (Salaj et al. 2021) [15]	Rec	4190	91.21
	ASRNN$_{\text{Nat. Mach. Intell., 2021}}$ (Yin, Corradi, and Bohté 2021) [1]	Rec	310	92.10
	RadLIF$_{\text{Front. Neurosci., 2022}}$ (Bittar and Garner 2022) [16]	Rec	1200	94.51
	DH-SNN$_{\text{Nat. Commun., 2024}}$ (Zheng et al. 2024) [9]	Rec	130	93.80
	TC-LIF$_{\text{AAAI, 2024}}$ (Zhang et al. 2024) [8]	Rec	200	94.84
	SPRiF (ours)	**Rec**	**130**	**94.55 ± 0.31**
SHD	Heterogeneous SNN$_{\text{Nat. Commun., 2021}}$ (Perez-Nieves et al. 2021) [3]	Rec	110	82.50
	ASRNN$_{\text{Nat. Mach. Intell., 2021}}$ (Yin, Corradi, and Bohté 2021) [1]	Rec	140	90.40
	DH-SNN$_{\text{Nat. Commun., 2024}}$ (Zheng et al. 2024) [9]	Rec	50	91.34
	BRF$_{\text{ICML, 2024}}$ (Higuchi et al. 2024) [6]	Rec	110	91.70 / 91.28 ±0.53*
	TC-LIF$_{\text{AAAI, 2024}}$ (Zhang et al. 2024) [8]	Rec	140	88.91 / 88.73 ±0.44*
	MPS-SNN$_{\text{ICLR, 2025}}$ (Ding et al. 2025) [29]	Rec	390	91.19
	DGN$_{\text{ICLR, 2026}}$ (Bai, Wang, and Yu 2026) [28]	Rec	220	87.78
	SPRiF (ours)	**Rec**	**50**	**91.66 ± 0.42**

Notes: K: thousands; bold: SPRiF. New accuracies: five-seed mean ± sample SD. Slashes: published/ours; equal parameter counts appear once. Published DMP-SNN variants: II on S-MNIST, I on PS-MNIST; our PS-MNIST rerun uses native architecture. GSC: validation; others: test.

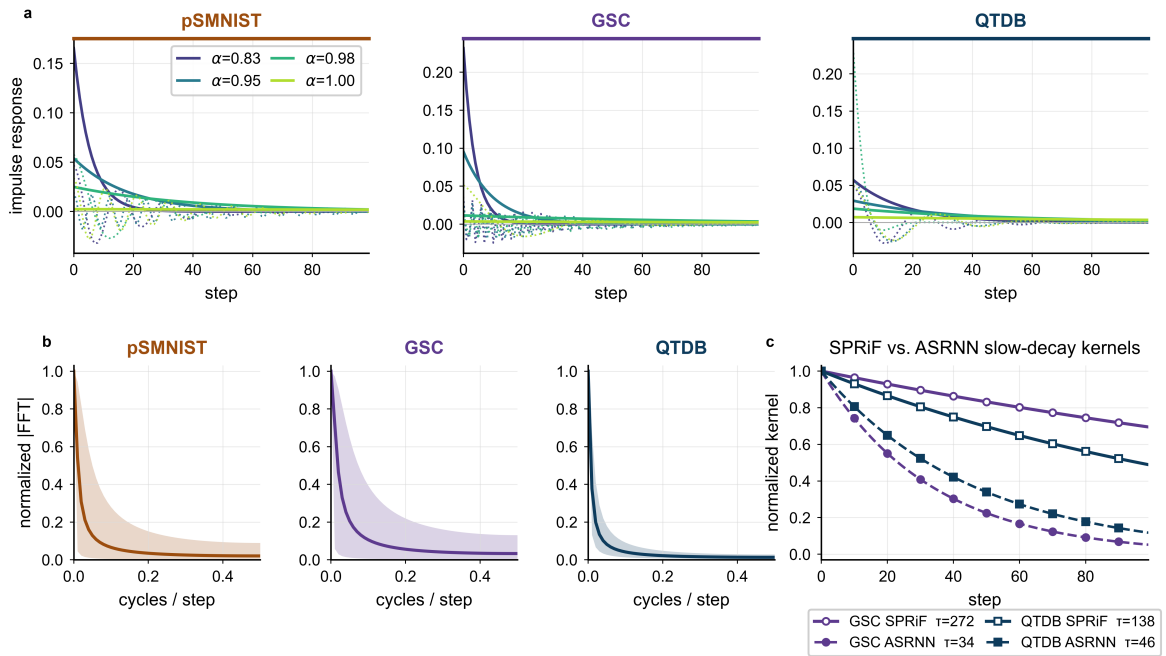


Fig. 2. Temporal responses across tasks. (a) Impulse responses on PS-MNIST, GSC and QTDB (left to right). (b) Normalized memory-state frequency responses for the same tasks; shading spans neurons, not training seeds. Frequency is in cycles per step. (c) Normalized slow-decay responses on GSC and QTDB: solid curves with open markers, SPRiF; dashed curves with filled markers, ASRNN.

TABLE II
STRUCTURAL ABLATIONS. THE THREE VARIANTS USE THE SAME LAYER WIDTHS, FIVE STATES AND ELEVEN LEARNED LOCAL COEFFICIENTS PER NEURON. ACCURACY IS MEAN ± SAMPLE SD (%) OVER FIVE SEEDS. THE SPRIF REFERENCE REUSES TABLE I.

Variant	PS-MNIST	SHD
SPRiF	95.86 ± 0.14	91.66 ± 0.42
MEMORY_RESET5	92.49 ± 0.36	87.10 ± 0.61
REAL_MODES3	93.52 ± 0.27	89.19 ± 0.50

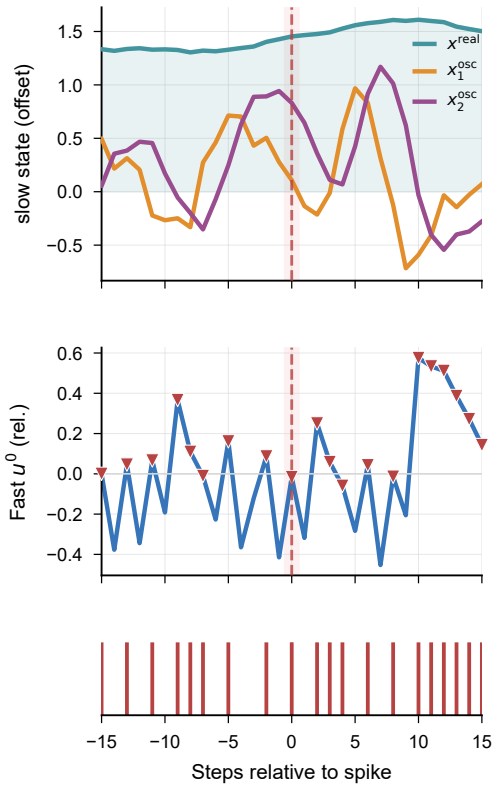


Fig. 3. SPRiF states around a recorded spike on PS-MNIST. From top to bottom: the real and two oscillatory memory coordinates (vertically offset), relative membrane voltage, and spikes. The dashed line marks the reference spike at relative step zero.

native architecture with layer-shared memory. Table I marks our accuracies with an asterisk and lists separate parameter counts only when network sizes differ.

SPRiF, MEMORY_RESET5 and REAL_MODES3 have the same layer widths, five states and eleven learned dynamics coefficients per neuron on PS-MNIST and SHD. We use the same partitions, network readout, training budget, checkpoint rule and seed-specific inputs across variants. The two comparisons alter different terms in the drive to the membrane.

MEMORY_RESET5. Compute the current spectral state $\mathbf{x}_t^-$, both readouts, the auxiliary state, and the membrane spike as in SPRiF. After firing, additionally update

$$\widehat{B} = \frac{B}{\|B\|_2}, \qquad \gamma_{\mathrm{mem}} = 1,$$
$$\mathbf{x}_t^+ = \mathbf{x}_t^- - \gamma_{\mathrm{mem}} \theta z_t \widehat{B}. \tag{29}$$

The next update starts from $\mathbf{x}_t^+$. Because h_t was computed from $\mathbf{x}_t^-$, the extra displacement enters the filter only at the following step. The reset has magnitude θ per spike and introduces no learned coefficient.

REAL_MODES3. Replace the spectral matrices by

$$A_{\mathrm{real}} = \mathrm{diag}(a_1, a_2, a_3), \qquad B_{\mathrm{real}} = \begin{bmatrix} 1 - a_1 \\ 1 - a_2 \\ 1 - a_3 \end{bmatrix}, \tag{30}$$

where $a_i = \sigma(a_{i,\mathrm{raw}})$ are independently learned. Three real decay parameters replace α, ρ, ω. The auxiliary filter, membrane, scalar reset, and six readout entries are retained.

The first real-mode time constant uses the τ_α interval, and the other two use independent draws from the τ_ρ interval. Each coordinate has a nonzero input coefficient. Simply setting the rotation angle to zero would leave one coordinate undriven from a zero initial state.

D. SI-DMS Task and Intervention Protocol

SI-DMS is a binary match/non-match task. Each trial presents two 100-ms cues separated by a delay, following a 50-ms baseline. Thirty input channels encode the left cue, right cue and background in groups of ten. At each 5-ms step, active cue channels emit a Bernoulli spike with probability 0.20 and background channels with probability 0.05. A recurrent layer of 64 neurons drives two leaky outputs with a 20-ms time constant. The final outputs determine the class.

The training delays are 200, 400, 800 and 1600 ms. Clean training uses natural firing throughout. Intervention training samples $K \in \{0, 1, 2, 4, 8\}$ forced-spike times during the delay. At each selected time, ten of the 64 units (nominally 15%) have their membrane candidates set to the current threshold plus 0.05. The induced spikes trigger each model's reset and enter the recurrent feedback. For SPRiF, the overwrite leaves the current spectral and auxiliary states intact. Training uses Adam with learning rate 0.003, batches of 256 and 2000 updates. Evaluation uses $K \in \{0, 1, 2, 4, 8, 16, 32, 40\}$ and adds a 2500-ms delay. We present the nominal 15% intervention condition, corresponding to ten of 64 neurons (15.625%) at each selected timestep.

For seed s, delay d, count K, and fraction f, the paired loss is

$$\Delta_{s,d}(K, f) = \mathrm{Acc}_{s,d}(0) - \mathrm{Acc}_{s,d}(K, f). \tag{31}$$

At $K = 40$ and nominal $f = 0.15$, we report the loss in percentage points over all five evaluation delays and separately at 2500 ms.

V. RESULTS AND DISCUSSION

A. Temporal Classification Performance

Table I summarizes five temporal benchmarks. On PS-MNIST, SPRiF reached $95.86 \pm 0.14\%$ with 67,000 parameters, compared with $95.41 \pm 0.52\%$ for the native DMP-SNN rerun at the same reported size. On SHD, SPRiF reached $91.66 \pm 0.42\%$ with 50,000 parameters. The BRF and TC-LIF reruns reached $91.28 \pm 0.53\%$ and $88.73 \pm 0.44\%$ with 110,000 and 140,000 parameters, respectively. The controls below test how SPRiF uses its states on these two tasks while keeping layer width and local capacity fixed.

TABLE III
SI-DMS ACCURACY (%, FIVE-SEED MEAN ± SAMPLE SD). STRESS: $K = 40$, TEN OF 64 NEURONS PER EVENT. Δ_{40}: MEAN $K = 0$ MINUS $K = 40$ ACCURACY (PERCENTAGE POINTS). DELAYS ARE EQUALLY WEIGHTED; 2500 MS IS UNSEEN DURING TRAINING.

		Five-delay average			2500 ms		
Training	Model	$K = 0$	$K = 40$	Δ_{40}	$K = 0$	$K = 40$	Δ_{40}
Clean	SPRiF	99.99 ± 0.01	99.42 ± 0.69	0.57	99.99 ± 0.01	99.99 ± 0.01	0.00
	MEMORY_RESET5	99.93 ± 0.07	94.74 ± 2.48	5.19	99.93 ± 0.10	99.84 ± 0.13	0.09
	REAL_MODES3	99.90 ± 0.20	95.37 ± 5.96	4.53	99.91 ± 0.16	99.87 ± 0.16	0.04
	ASRNN	99.94 ± 0.03	48.56 ± 1.99	51.39	99.89 ± 0.13	47.50 ± 4.48	52.39
	BRF	82.00 ± 0.76	54.80 ± 1.65	27.19	61.87 ± 4.95	50.62 ± 0.97	11.25
	LIF	88.46 ± 18.87	51.01 ± 1.56	37.44	83.35 ± 20.95	53.36 ± 7.55	30.00
Intervention	SPRiF	99.69 ± 0.67	99.33 ± 0.78	0.36	99.65 ± 0.71	99.65 ± 0.72	0.00
	MEMORY_RESET5	94.46 ± 12.23	88.54 ± 20.25	5.92	93.81 ± 13.75	91.23 ± 19.32	2.57
	REAL_MODES3	89.94 ± 22.47	89.91 ± 22.45	0.03	90.08 ± 22.16	90.04 ± 22.13	0.04
	ASRNN	53.22 ± 3.17	49.85 ± 0.13	3.37	49.94 ± 0.26	49.80 ± 0.77	0.13
	BRF	73.28 ± 3.42	51.34 ± 0.48	21.94	57.15 ± 7.99	50.11 ± 0.40	7.04
	LIF	56.53 ± 1.42	49.98 ± 0.24	6.55	50.09 ± 0.61	50.07 ± 0.83	0.02

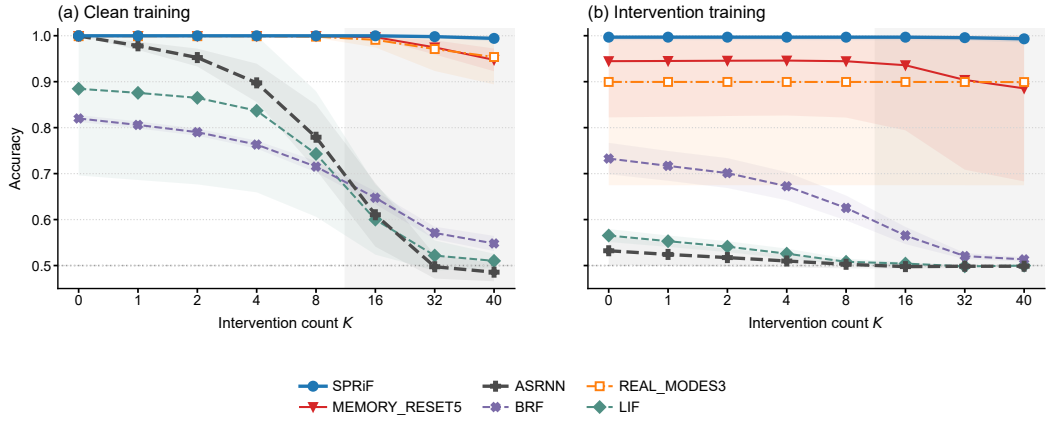


Fig. 4. SI-DMS accuracy after (a) clean and (b) intervention training (ten of 64 neurons forced per event). Lines: five-seed means with equally weighted delays; bands: ± sample SD, clipped to $[0, 1]$. Dotted: 50% chance; gray: counts beyond training ($K > 8$). Conditions are equally spaced.

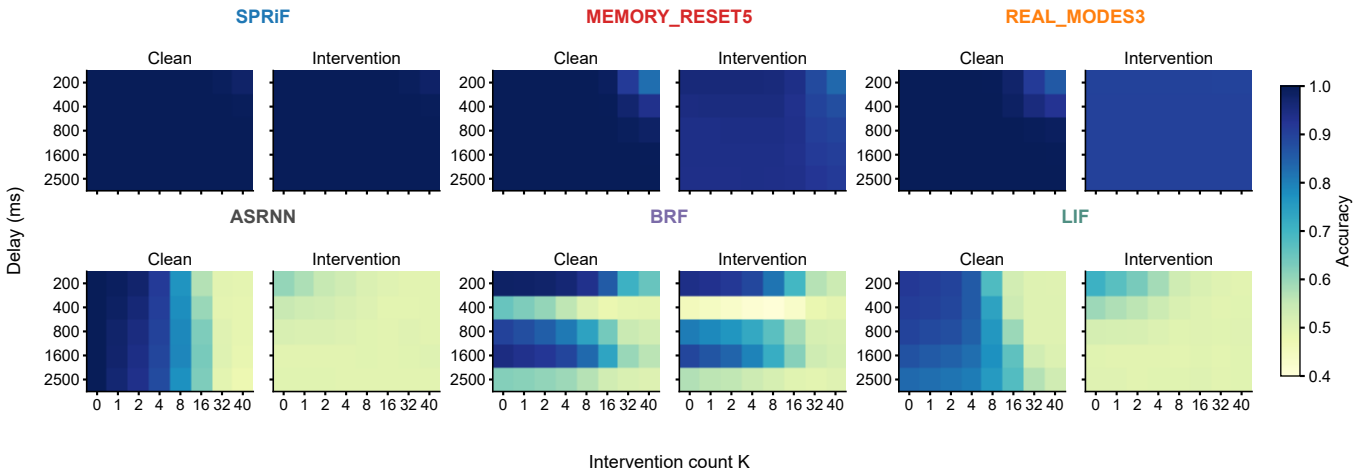


Fig. 5. SI-DMS accuracy by delay and intervention count after Clean and Intervention training (five-seed means; shared scale, 40%–100%). Each event targets ten of 64 neurons; $K = 0$ is unforced. The 2500-ms delay and $K = 16, 32, 40$ are unseen during intervention training.

B. Mechanistic Analysis and Ablation Studies

Figure 2 connects the spectral basis in Eq. (11) to the temporal responses on PS-MNIST, GSC and QTDB. The impulse responses show a real decay alongside damped oscillatory components (Fig. 2a). These components provide traces with different durations and phases for the membrane projections in Eq. (14). The frequency responses summarize the memory

states across neurons in each task (Fig. 2b). Their shaded ranges describe variation across neurons. In the slow-decay comparison, the displayed SPRiF responses retain a larger normalized amplitude than ASRNN over the plotted interval on GSC and QTDB (Fig. 2c).

Figure 3 shows SPRiF memory states, membrane voltage and spikes around a recorded firing event on PS-MNIST. The memory trajectories continue through the marked event, while the membrane trace reflects firing and reset. This illustrates the separation in Eq. (4): the spike changes the stored membrane voltage without subtracting from the spectral memory.

The two neurons have the same scalar membrane reset. In MEMORY_RESET5, a spike at t also contributes $r[1] = -\theta \mathbf{g}_0^\top A\widehat{B}$ to the drive at $t+1$ (Eq. (24)). The filter first carries part of that displacement to the membrane at $t+2$. Its sign depends on the learned projections. A memory reset thus adds a spike-dependent signal to the readout even if the incoming current is held fixed. SPRiF omits this path and scored 3.37 percentage points higher on PS-MNIST and 4.56 points higher on SHD (Table II).

The mode comparison keeps the reset on the membrane. In REAL_MODES3, the readout combines three driven exponential decays. In SPRiF, an input also leaves a trace whose direction rotates with elapsed time. Its projection can therefore weight successive phases differently under a common decay envelope (Eq. (11)). The real-plus-rotational block scored 2.34 percentage points higher on PS-MNIST and 2.47 higher on SHD than the three-real-mode block. This comparison tests the choice of basis and its associated input coupling together.

C. Robustness to Spike Interventions

After clean training, SPRiF accuracy decreased from 99.99% at $K = 0$ to 99.42% at $K = 40$ (Fig. 4). At $K = 40$, MEMORY_RESET5 and REAL_MODES3 reached 94.74% and 95.37%, while ASRNN, BRF and LIF ranged from 48.56% to 54.80% (Table III).

After intervention training, SPRiF retained $99.33 \pm 0.78\%$ at $K = 40$, a decrease of 0.36 percentage points from $K = 0$. The matched controls had lower stressed accuracy. In particular, REAL_MODES3 showed a near-flat mean curve alongside a sample SD of 22.45 percentage points, underscoring variation across seeds.

The heatmaps show that SPRiF's mean accuracy exceeded 97.7% at every displayed delay and intervention count. At the unseen 2500-ms delay and $K = 40$, SPRiF reached 99.99% after clean and 99.65% after intervention training (Fig. 5). Clean-trained controls lost more accuracy at shorter delays.

VI. Conclusion

SPRiF gives the membrane a decaying trace and a phase-dependent trace of past input, each available through an immediate and a delayed projection. Each spike resets the membrane and can re-enter the spectral pathway one step later through recurrent current. Extending reset to spectral memory adds a distinct spike-history term to the drive. At fixed local capacity, both preserving that memory at discharge and retaining its rotational pair improved mean accuracy on PS-MNIST and SHD. On SI-DMS, SPRiF sustained accurate cue comparison under both training regimes at an unseen delay and intervention counts beyond training.

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